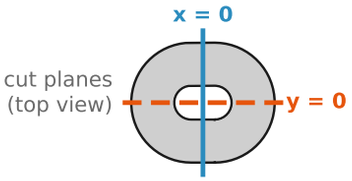
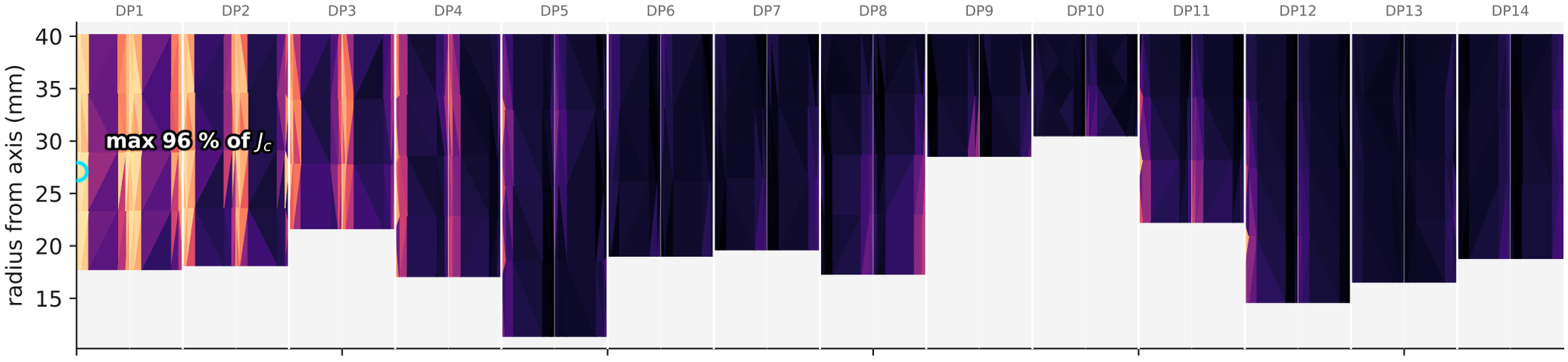
Local critical-current loading of the optimised 28-layer REBCO racetrack coil

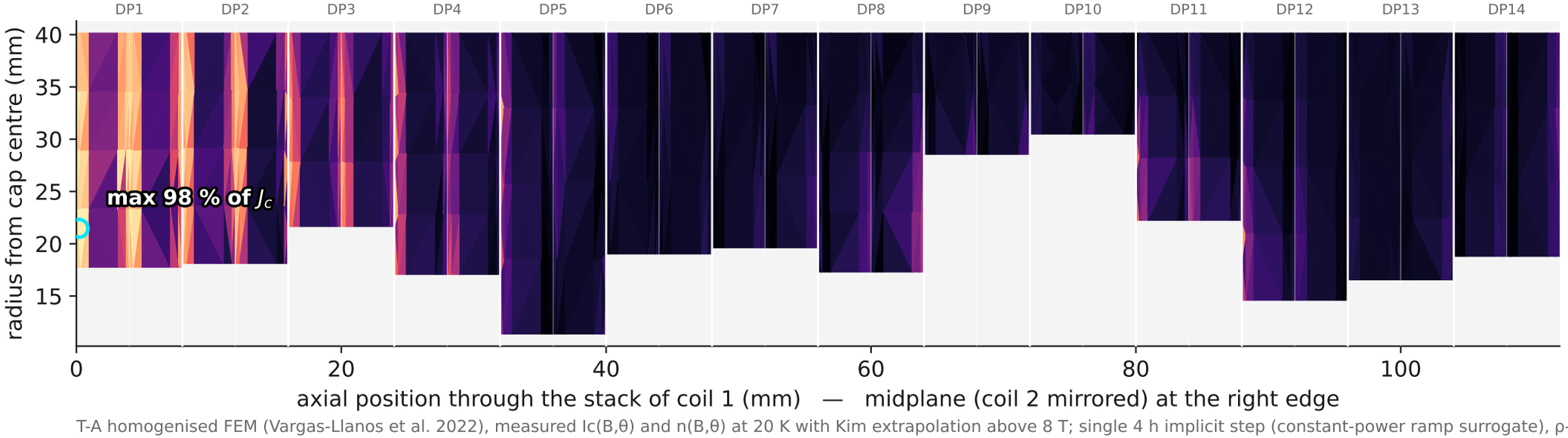
131.9 A per turn, at the end of a 2 h constant-power ramp · peak 99 % of J_c · 0 of 29,364 coil cells above J_c

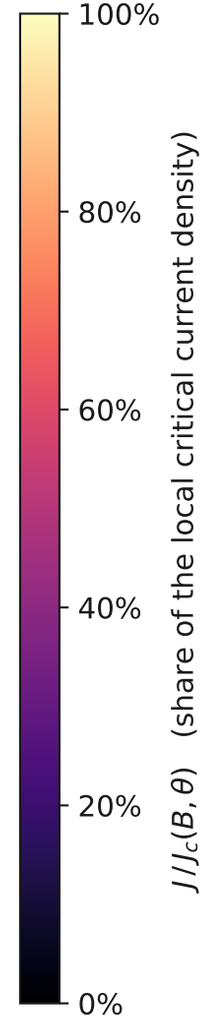


Straight-leg cross-section (x = 0)



End-cap cross-section (y = 0, through the cap tip)





T-A homogenised FEM (Vargas-Llanos et al. 2022), measured Ic(B,θ) and n(B,θ) at 20 K with Kim extrapolation above 8 T; single 4 h implicit step (constant-power ramp surrogate), ρ-floor 0.9.
Each triangle is one finite-element face on the symmetry plane, coloured by its cell value (no interpolation); in-plane element size ≈ 7 mm.
3 degenerate (sliver) element(s), shape quality < 0.001, excluded from the peak and the count.
DP = double pancake, numbered from the axial end of the coil to the midplane.